



Kotlin

Training Assignment

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RECORD OF CHANGES

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Assignment 1: Basic OOP in Kotlin

Create a Vehicle class which is an abstract class that represents the means of transport. Define attributes such as license plate, year of manufacture, and price. Provides a constructor to initialize properties and methods to access information and calculate the total price.

Create two subclasses Car and Motorcycle that inherit from Vehicle class. Define specific properties for each vehicle type, along with the corresponding constructor and methods.

Create a VehicleManager class to manage the list of vehicles. This class will have a list of Vehicle objects and provide methods to add vehicles, display the list, and calculate the total price.

In the main function or a similar method, create a VehicleManager object and do some operations that add vehicles, display a list, and calculate the total price.

Technical Requirements:

- Kotlin OOP(using abstract class or interface)

Assignment 2: Advanced OOP in Kotlin

Build a contact management application. Each contact can contain many types of contact information such as phone numbers, email addresses, social media accounts

Technical Requirements:

Generics and Sealed Classes

- Create a Sealed Class ContactInformation to represent the type of contact information.
- Create subclasses of ContactInformation like PhoneNumber, EmailAddress, SocialMediaAccount, etc.
- Use Generics to create a Contact<T: ContactInformation> class that represents a phonebook with specific contact information. This class will have a property named list T.

Object Expressions and Object Declarations:

- Use Object Expressions to create a sample directory with some contact information.
- Create an Object Declaration as ContactManager to manage contacts.
- In ContactManager, use the sample contacts created with Object Expressions.

Program implementation:

- In the main function, use ContactManager to add and display contacts.
- Displays a list of contacts and contact information for each contact.

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